

# ACHYUT SRIDHAR KULKARNI

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*ML Systems Engineer with 4+ years building low-latency recommendation systems, real-time CV pipelines, and production LLM applications end-to-end, from data ingestion and ranking architecture to scalable deployment. Known for finding what breaks before it does: contributed XAI audit findings to Toyota Research Institute's AV safety playbooks after diagnosing critical false-braking failure modes in a self-driving perception pipeline.*

## PROFESSIONAL EXPERIENCE

**Data Scientist (Machine Learning) | Stealth Startup (Event Discovery Platform) | Charlotte, NC (Remote)** Aug 2025 - Present

- Architected an event ingestion pipeline to power city-wide event discovery in Nashville, centralizing fragmented event listings across heterogeneous web sources and targeting 50% expansion of discoverable local event inventory.
- Drove implementation of Neo4j-based recommendation system that ranked events and users leveraging graph and location signals, enabling personalized event discovery and improving recommendation relevance by 28%.
- Engineered low-latency data pipelines with optimized query patterns and caching layers, reducing latency 40% under real-time load.
- Translated model outputs into product-facing ranking strategies, reducing churn 15% via improved discovery pathways.
- Implemented and deployed multi-stage ranking pipeline (candidate generation + re-ranking) optimizing precision-recall tradeoffs in production.

**Data and Analytics Engineer | DigiCXO Partners | Bengaluru, India** Mar 2023 - Aug 2023

- Constructed automated ETL pipelines integrating 5+ heterogeneous data sources, reducing manual preprocessing time 30%.
- Designed data models and dashboards for executive KPI tracking, eliminating more than 10 hours of weekly manual reporting.

**Software Engineer (AI/ML) | Megh Computing | Bengaluru, India** Aug 2020 - Mar 2023

- Productionized real-time video analytics pipelines at scale across smart city, retail and finance domains using CPU, GPU and FPGA acceleration.
- Reduced edge inference latency 40% by optimizing YOLOv5/Faster R-CNN using TensorRT, OpenVINO and model pruning techniques.
- Operationalized cloud inference systems enabling reliable deployment in low-bandwidth environments, improving operational efficiency 30%.
- Optimized models against client-defined KPIs and latency SLAs, accelerating production deployment and reducing iteration cycles.

## EDUCATION

**M.S. Data Science | Rochester Institute of Technology, Rochester, NY** GPA 3.9 / 4.0 | Aug 2023 - May 2025

**B.E. Information Science | BNM Institute of Technology, Bengaluru, India** GPA 3.4 / 4.0 | Aug 2016 - Aug 2020

## TECHNICAL SKILLS

**Interpretability & Evaluation:** TCAV, CRAFT, LIME, SHAP, NMF Concept Extraction, A/B Testing, Causal Inference, MMLU, TruthfulQA

**ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, Pandas, NumPy

**Recommender Systems & Graph ML:** Neo4j, Graph Embeddings, Candidate Generation, Vector Search, Ranking Systems

**Computer Vision:** YOLOv5, Faster R-CNN, TensorRT, OpenVINO, Object Detection, Multi-Object Tracking, Edge AI, Optical Character Recognition

**LLMs & GenAI:** Transformers, RAG, LoRA/PEFT Fine-tuning, Prompt Engineering, LangChain, LangGraph

**Data and MLOps:** Spark, Airflow, PostgreSQL, MongoDB, Redis, AWS (SageMaker, S3), GCP (Vertex AI, BigQuery), Docker, CI/CD, MLflow

**Languages:** Python, SQL, C++, R

## SELECTED PROJECTS

**Interpretable AV Perception (M.S. Capstone, RIT 2024-2025) | PyTorch, TCAV, CRAFT, YOLOv3, OpenCV | w/ Toyota Research Institute (TRI), University of Florida, UC Irvine | [View on GitHub](#)**

- Diagnosed false-braking failure modes using TCAV and CRAFT concept-level interpretability on a self-driving perception pipeline; targeted retraining reduced false positives 31% and improved pedestrian detection accuracy 18%, with findings integrated into XAI audit playbooks at TRI.

**Gemma4 Video Realism Audit | Gemma 4, Transformers, OpenCV, Multimodal Evaluation | Kaggle Gemma 4 Good Hackathon | [View on GitHub](#)**

- Built a Gemma 4 multimodal auditing system for AI-generated video realism: extracted representative frames, detected synthesis artifacts, and ensembled repeated model judgments via consensus voting to rank synthetic outputs by visual convincingness.

**WebPage Builder AI | LLM Agents, Prompt Engineering, Ollama | [View on GitHub](#)**

- Shipped a local agentic system that converts natural language prompts into validated, browser-ready web prototypes using constrained code generation, automated critique-and-repair loops and commit-gated validation eliminating manual iteration between prompt and result.

**MedGemma Edge CT Triage Pipeline | MedGemma 4B, Hugging Face | Kaggle MedGemma Impact Challenge | [View on GitHub](#)**

- Developed an end-to-end chest CT triage pipeline using MedGemma 4B with HU windowing, structured clinical prompts, controlled vocabulary normalization and scan-level urgency aggregation applying rigorous LLM evaluation methodology to critical diagnostic prioritization tasks.

## RECOGNITION & CERTIFICATIONS

**Winner - SCB Business Analytics Competition 2025 :** Benchmarked GPT-4, Claude 3 and Mistral on MMLU, TruthfulQA and ARC; built governance rubrics for fairness, compliance and ROI; recommended Claude 3 as RIT's institutional AI platform. Ranked first. [Link](#)

**Oracle Cloud Infrastructure AI Certifications (2025) :** Generative AI Professional, AI Foundations Associate, AI Vector Search Professional, AI Data Science Professional - validated through production RAG and vector search implementations.

**Published Research, IJLTET 2020 :** "Application to Detect Skin Cancer using CNN" - MobileNet, optimised for sub-\$50 edge deployment with explainable diagnostic outputs. [Read paper](#)